

VOLUME I

— *Ten Cities, 200 Years* —

OHIO

A generative cartography of population, place, and change.

Census curves • real geography • pen-plotted originals

project-artinscience

proof edition — June 2026

STD

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Each chapter: one plotted plate, then forecast methods, data sources, and notes.

OHIO

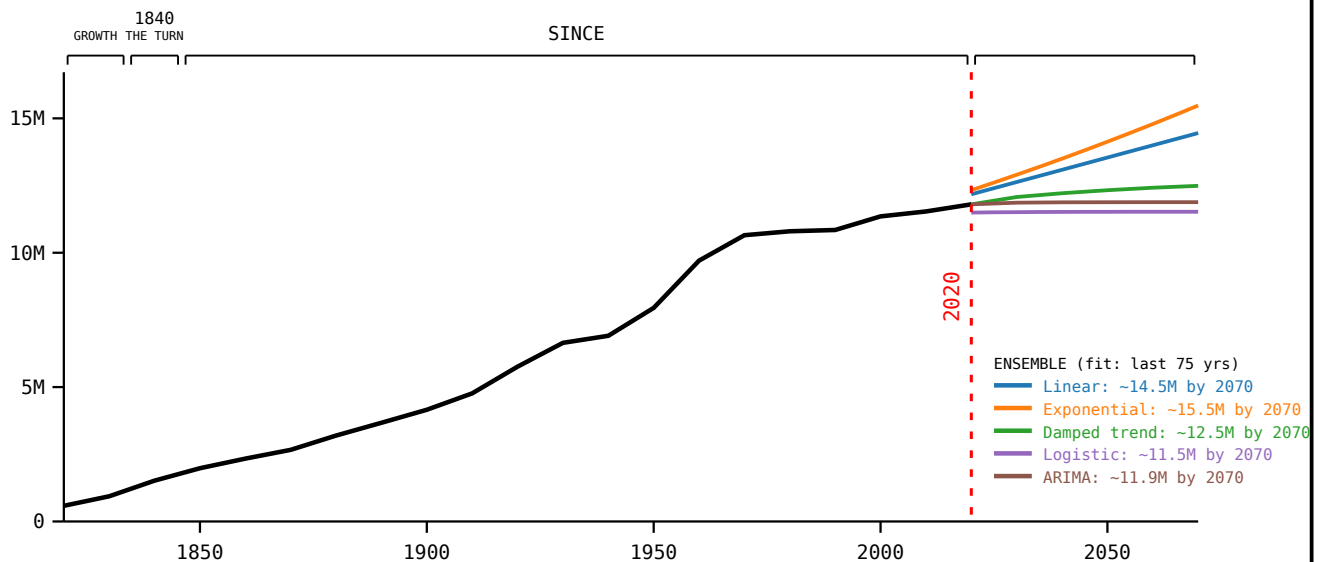
Chapter 1 • 1820–2020, projected to 2120



Before the ten cities, the state itself: two centuries of Ohio in a single line, climbing through canal, rail, oil, rubber and steel, then flattening after 1970. The plateau is an average — and an average hides everything.



LEGEND — State outline — Counties /// Ohio (this state)



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VOLUME I — OHIO: TEN CITIES, 200 YEARS

Ohio

Chapter 1 — Methodology: Forecast Algorithms

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1. Linear — ordinary least squares. A straight line through the recent points: constant absolute change per year — the conservative baseline.

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VOLUME I — OHIO: TEN CITIES, 200 YEARS

Ohio

Chapter 1 — Notes & Topics for Discussion (draft)

If you could time-travel, the year to visit

Also notable for

Population — detail, density & trend

Race & ethnicity ratios

Age & gender ratios

Tourism & attractions

Businesses & major employers

Crime & safety

Schools & education

Land use — residential / commercial / industrial zones

Nearby cities, towns & natural areas

Placeholders — to be researched and written in a later pass.

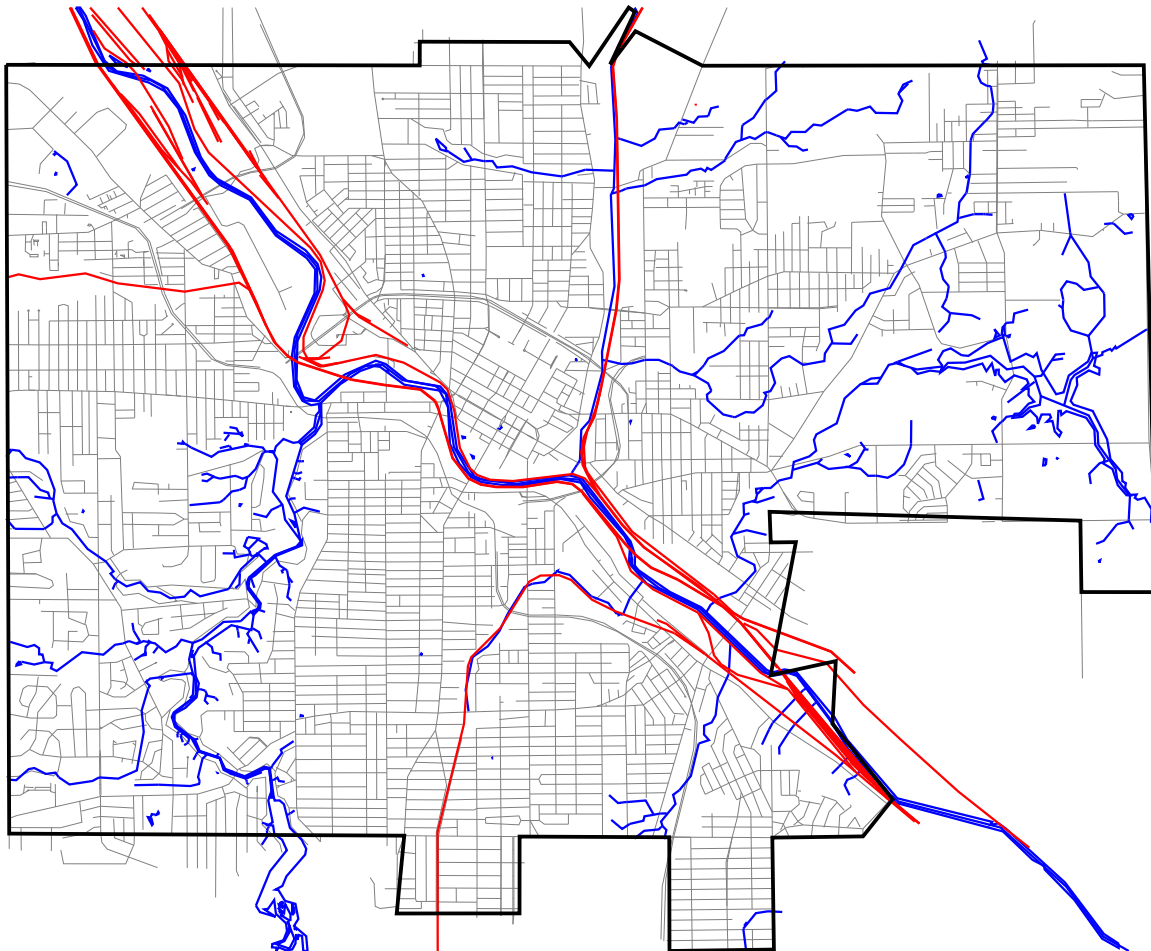
YOUNGSTOWN

Chapter 2 • 1820–2020, projected to 2120

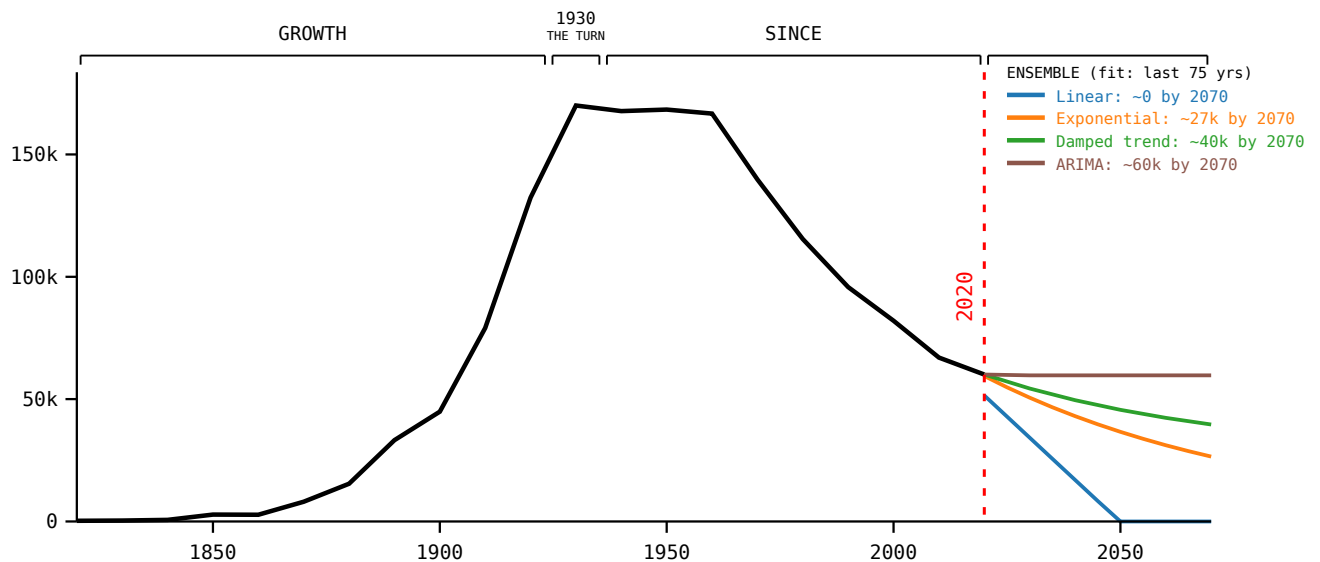


MAHONING CO.

Steel built this city on the Mahoning River; on Black Monday, 19 September 1977, the closures began and a century-long rise turned to the sharpest fall in the volume.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



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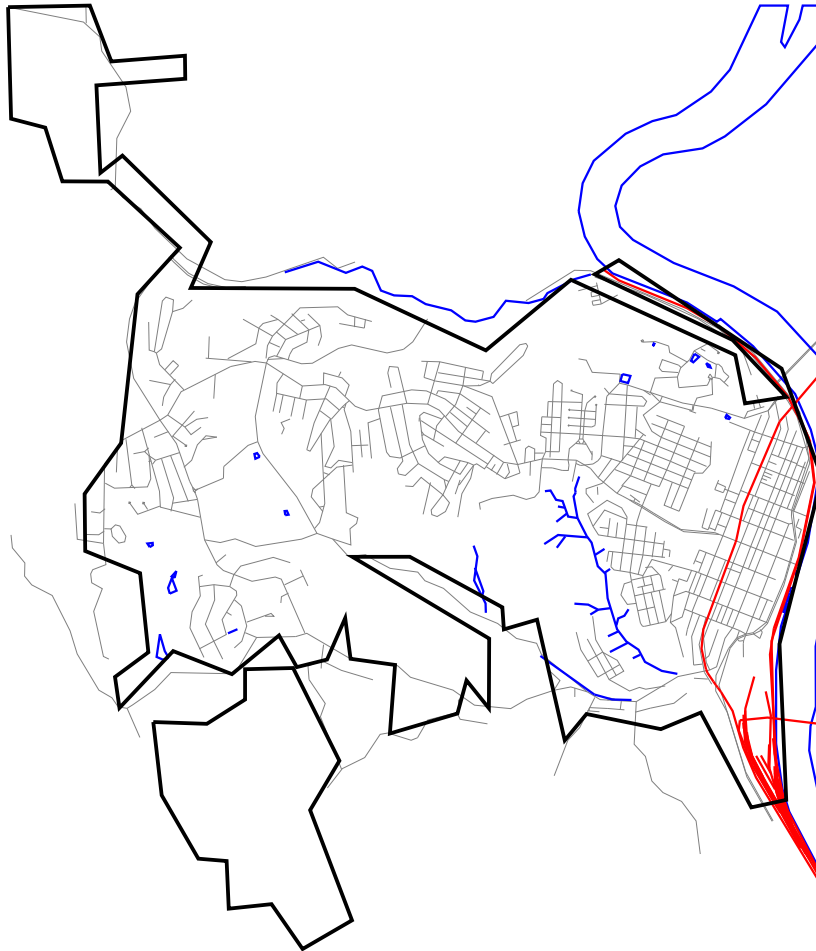
STEUBENVILLE

Chapter 3 • 1820–2020, projected to 2120

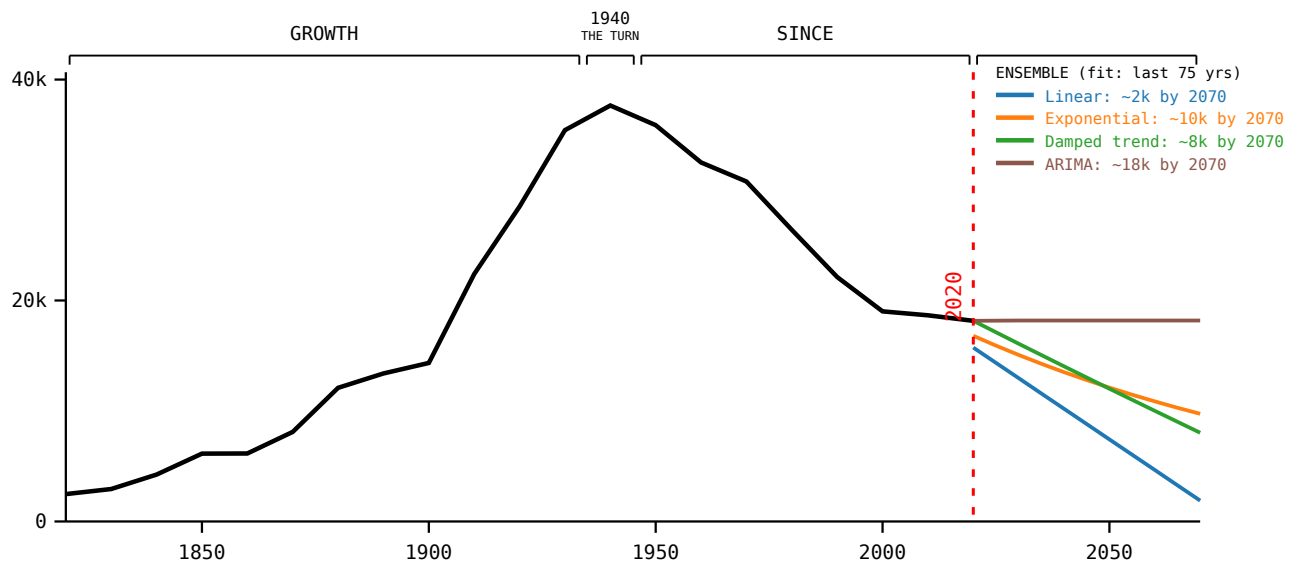


JEFFERSON CO.

An Ohio River town of steel and coal that peaked in 1940 and has faded slowly ever since; the boyhood home of Dean Martin.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



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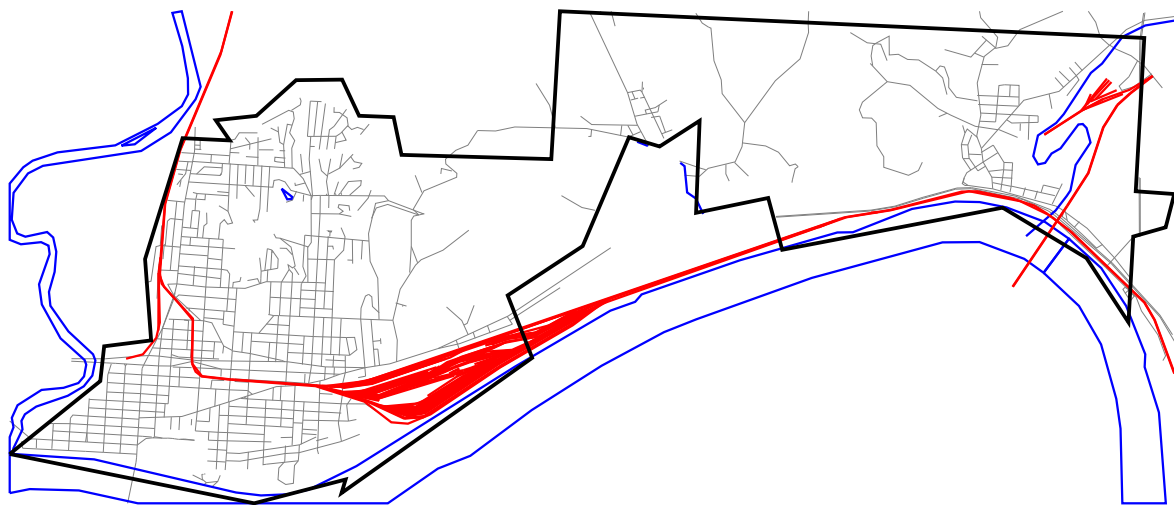
PORTSMOUTH

Chapter 4 • 1820–2020, projected to 2120

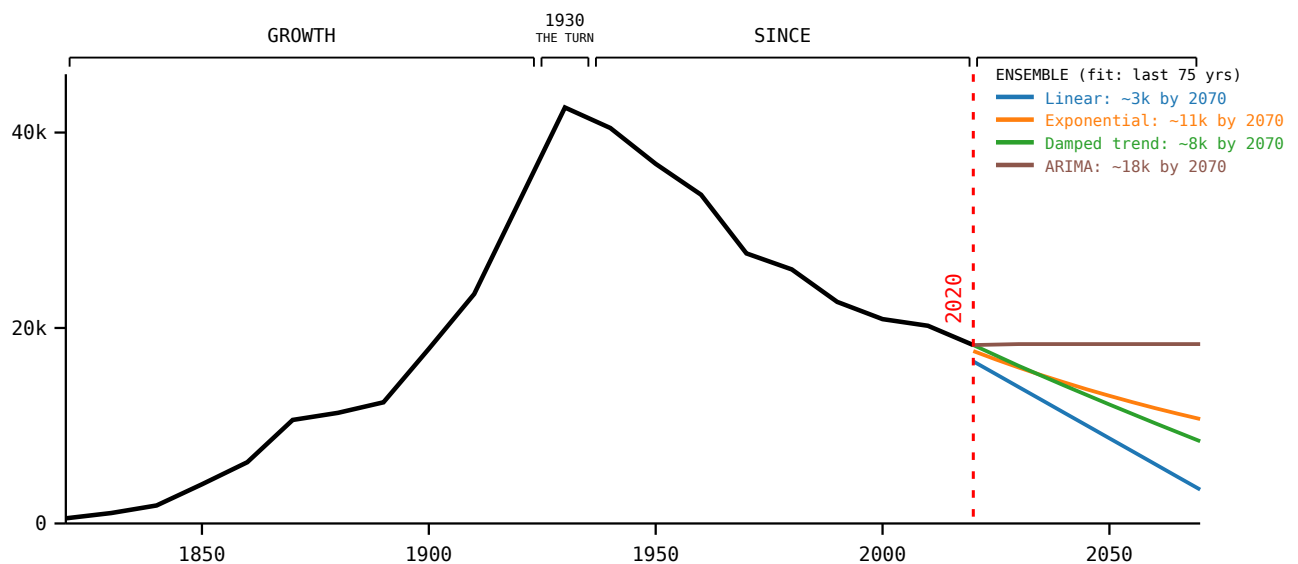


SCIOTO CO.

Steel, shoes and firebrick carried Portsmouth to a 1930 peak; the catastrophic 1937 flood marks where its long decline begins.



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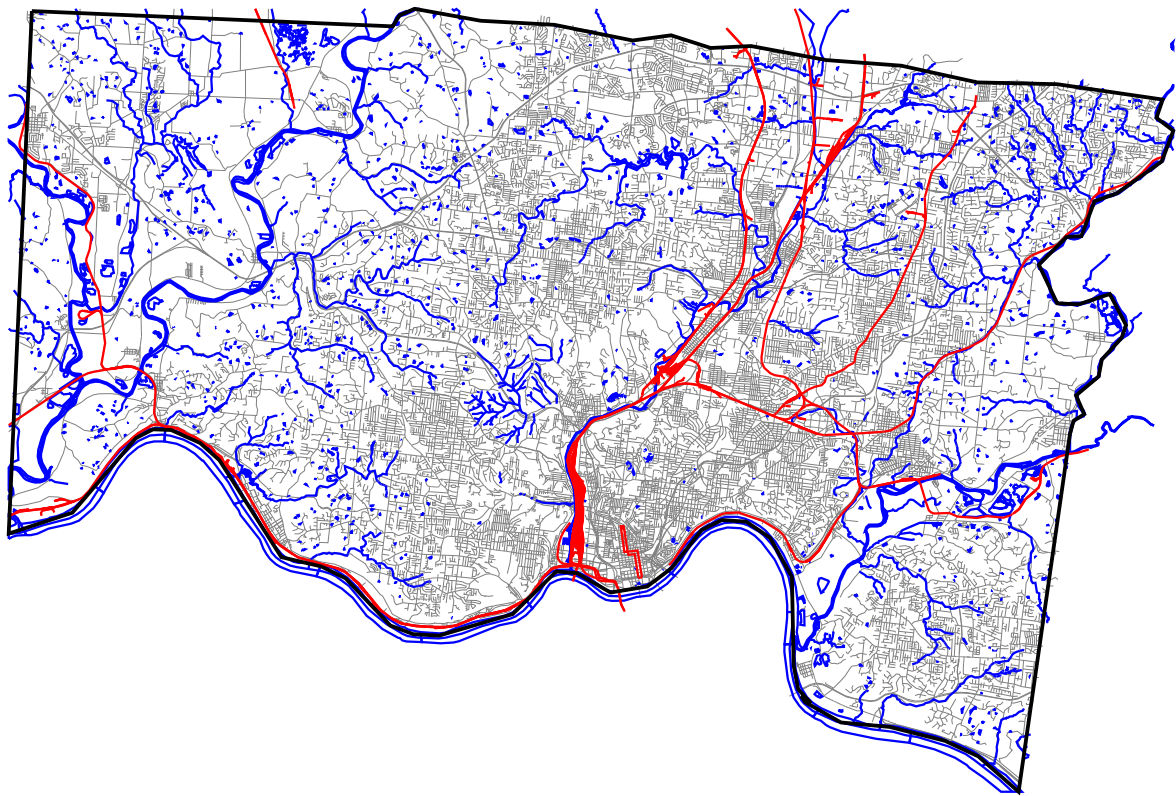
HAMILTON

Chapter 5 • 1820–2020, projected to 2120

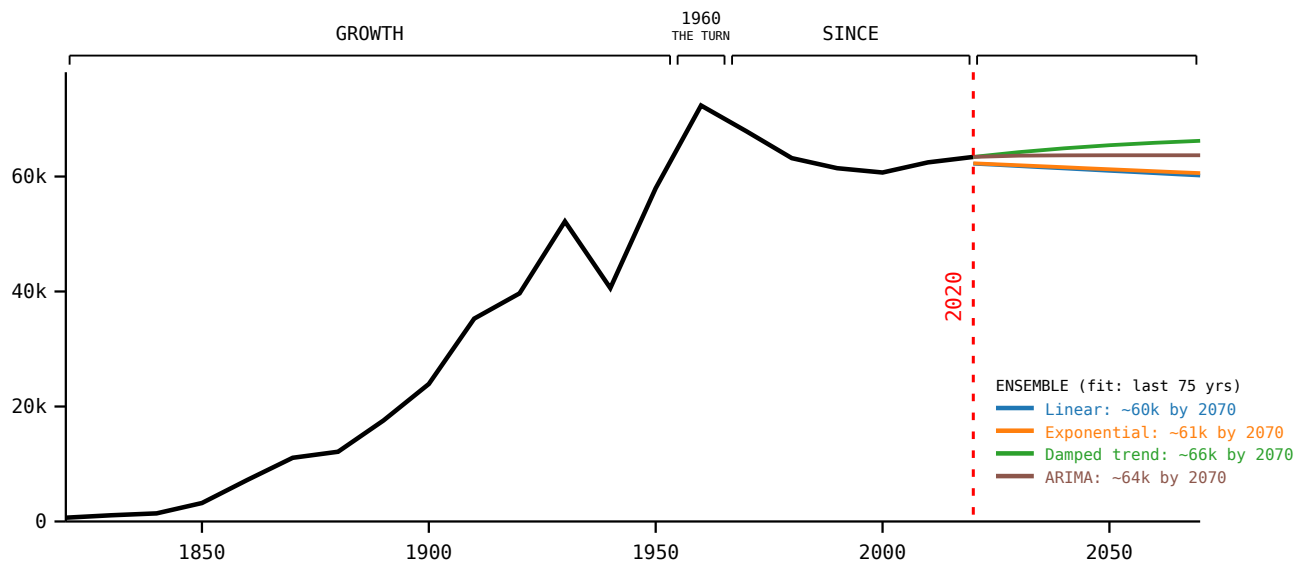


BUTLER CO.

A paper and safe-making town that punctuated itself into “Hamilton!” in 1986, eroded through the 1990s, then clawed its way back.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



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VOLUME I — OHIO: TEN CITIES, 200 YEARS

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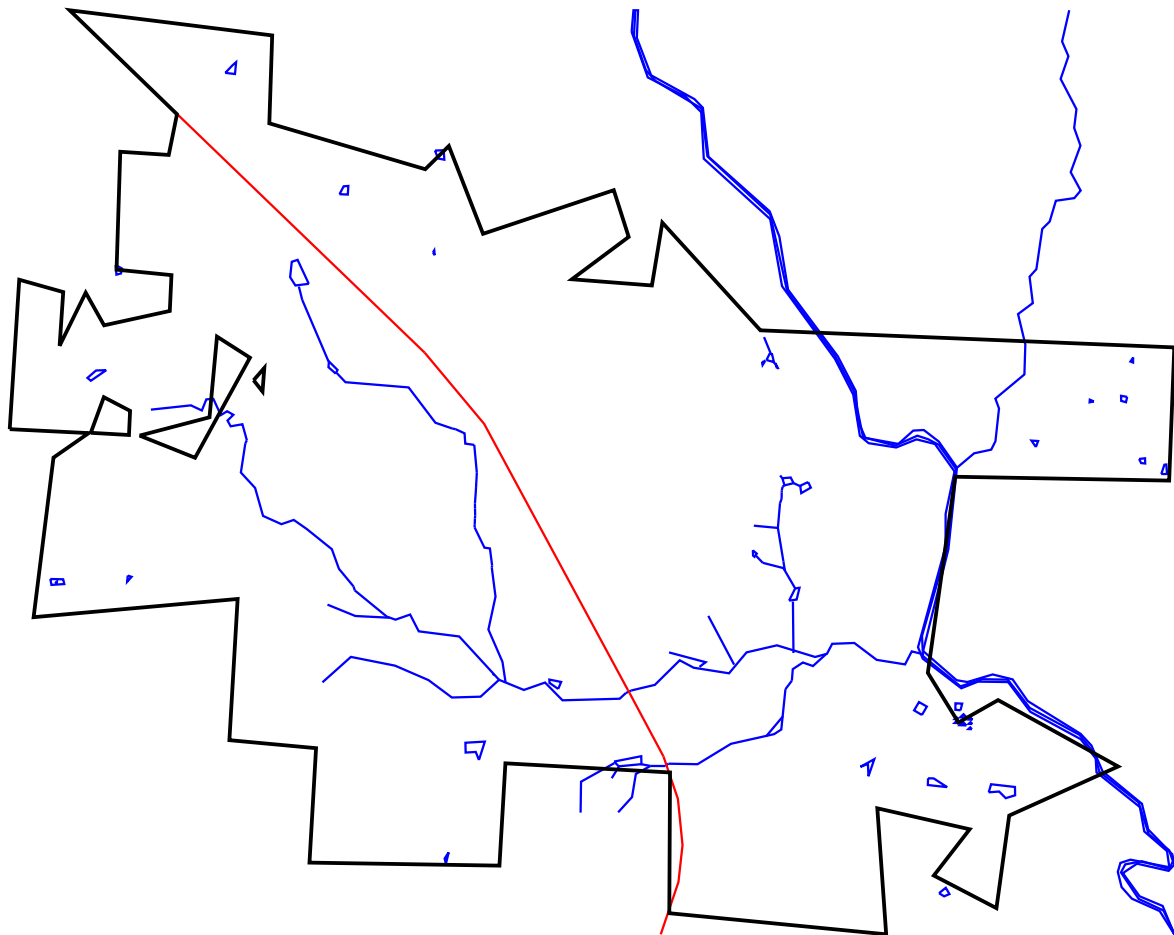
OXFORD

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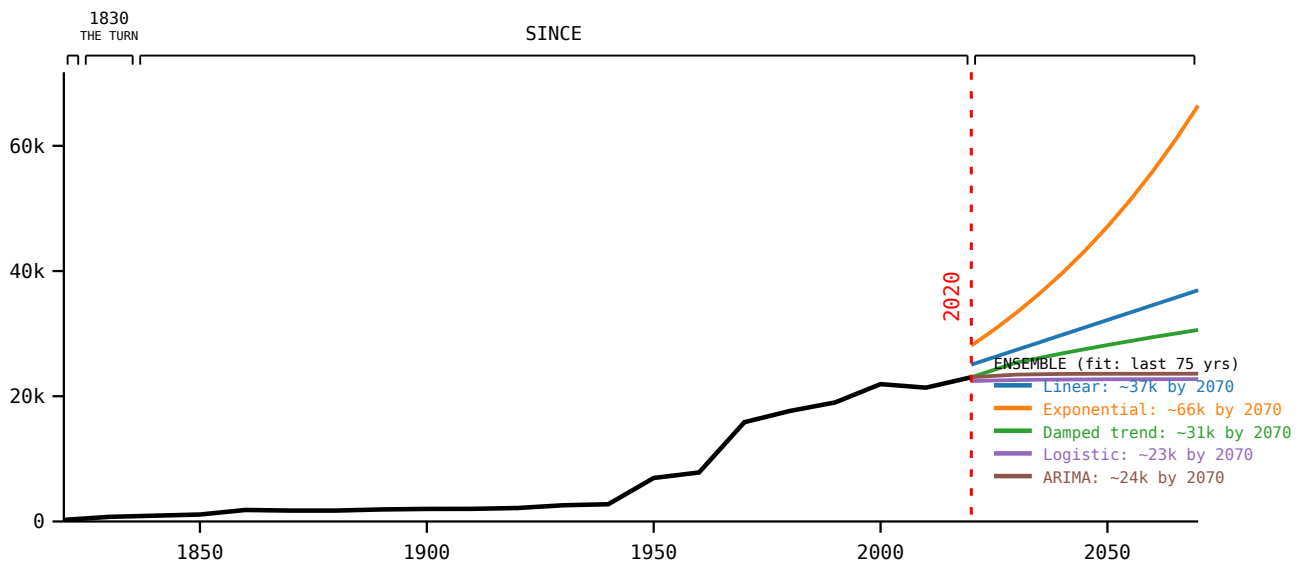


BUTLER CO.

Not a boom-and-bust but a patient climb tied to Miami University, chartered in 1809 and opened in 1824.



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VOLUME I — OHIO: TEN CITIES, 200 YEARS

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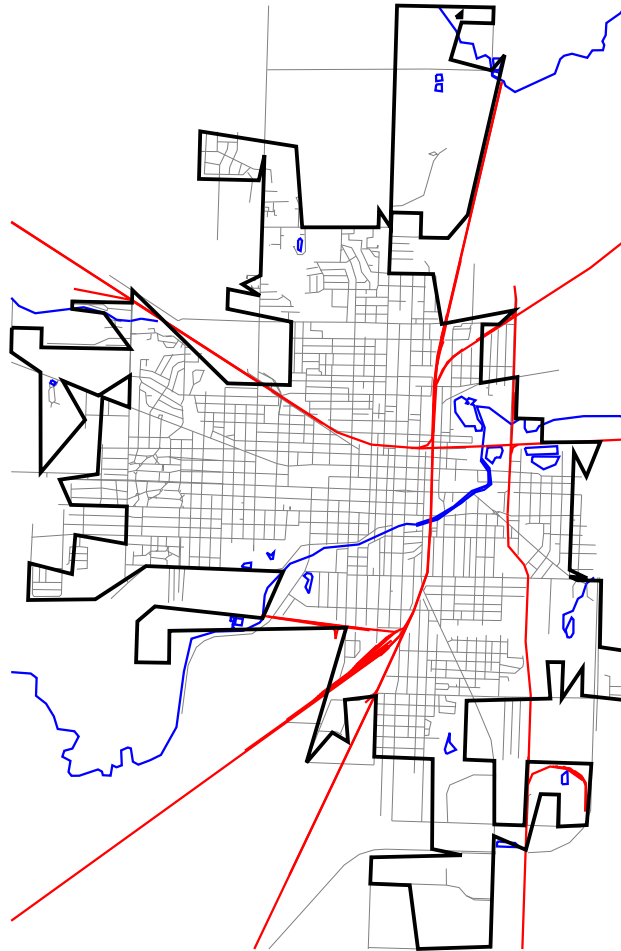
LIMA

Chapter 7 • 1820–2020, projected to 2120

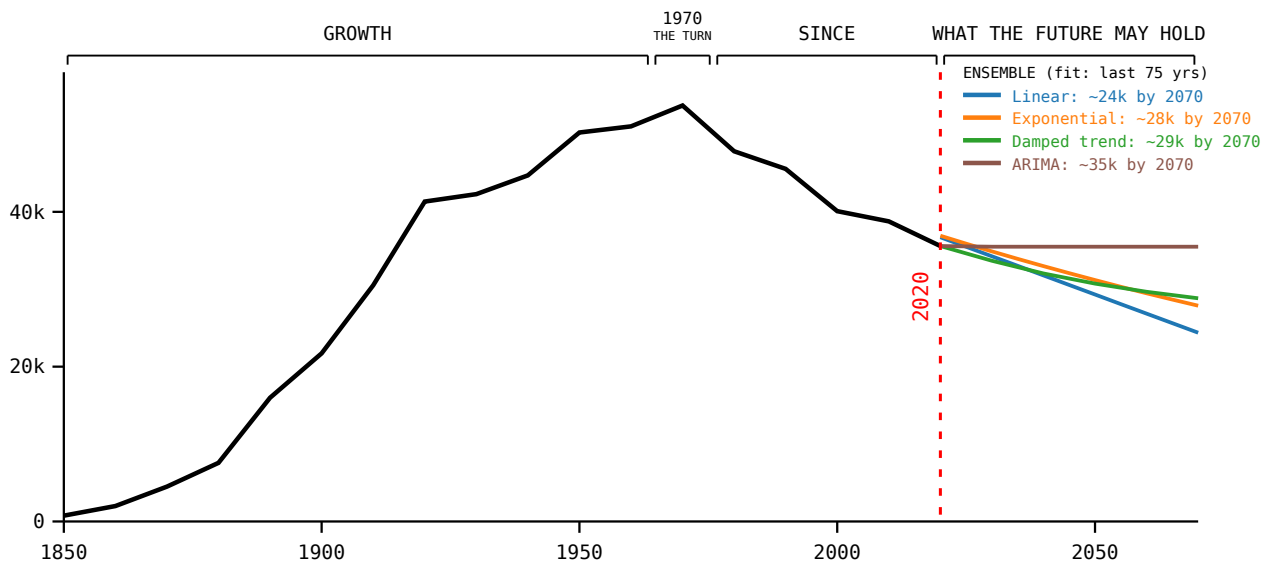


ALLEN CO.

Oil struck in 1885 and the Lima Locomotive Works built “Super-Power” steam; the city peaked around 1970 and eroded by a third.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



project-artinscience • seed 1970 • edition _____ / _____ • signed _____

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Lima

Chapter 7 — Methodology: Forecast Algorithms

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VOLUME I — OHIO: TEN CITIES, 200 YEARS

Lima

Chapter 7 — Data Sources & Software

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STD

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Lima

Chapter 7 — Notes & Topics for Discussion (draft)

If you could time-travel, the year to visit

Also notable for

Population — detail, density & trend

Race & ethnicity ratios

Age & gender ratios

Tourism & attractions

Businesses & major employers

Crime & safety

Schools & education

Land use — residential / commercial / industrial zones

Nearby cities, towns & natural areas

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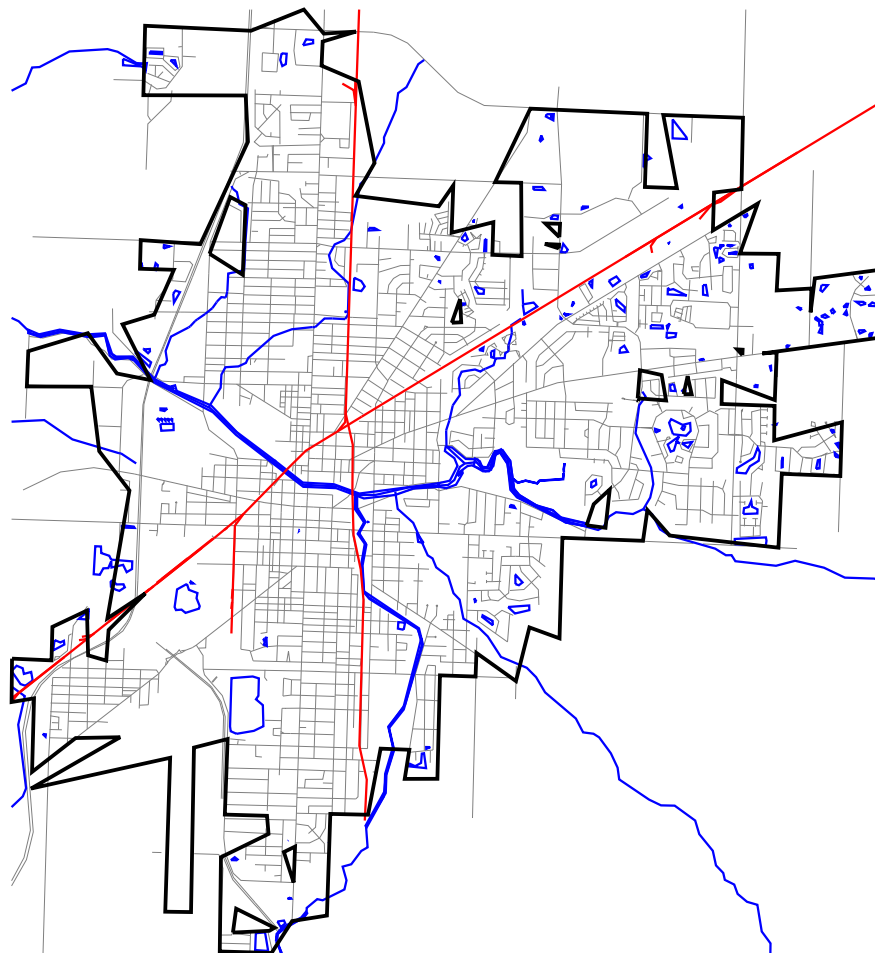
FINDLAY

Chapter 8 • 1820–2020, projected to 2120



HANCOCK CO.

An 1880s natural-gas boom made it the “City of Light”; unlike the steel towns, its curve never truly crashes.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



project-artinscience • seed 1840 • edition _____ / _____ • signed _____

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Findlay

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Findlay

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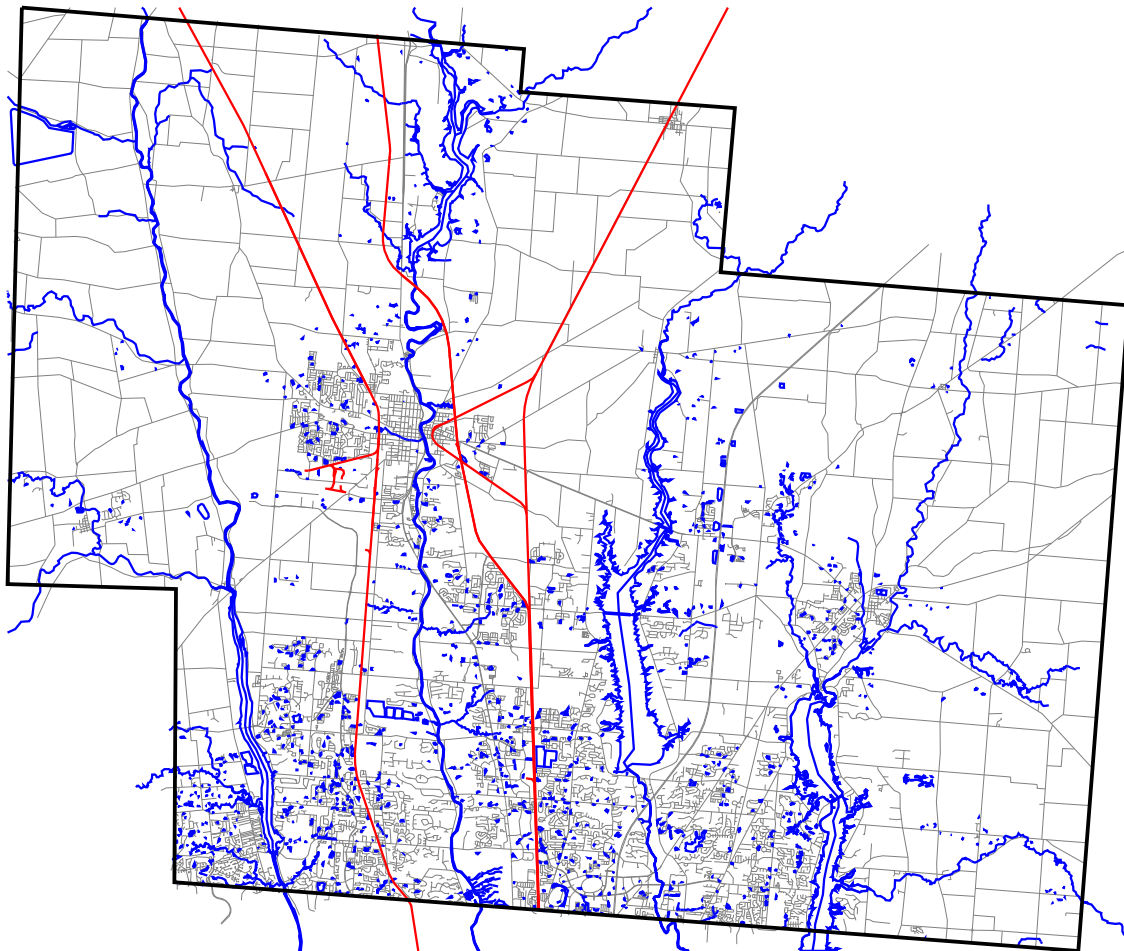
DELAWARE

Chapter 9 • 1820–2020, projected to 2120

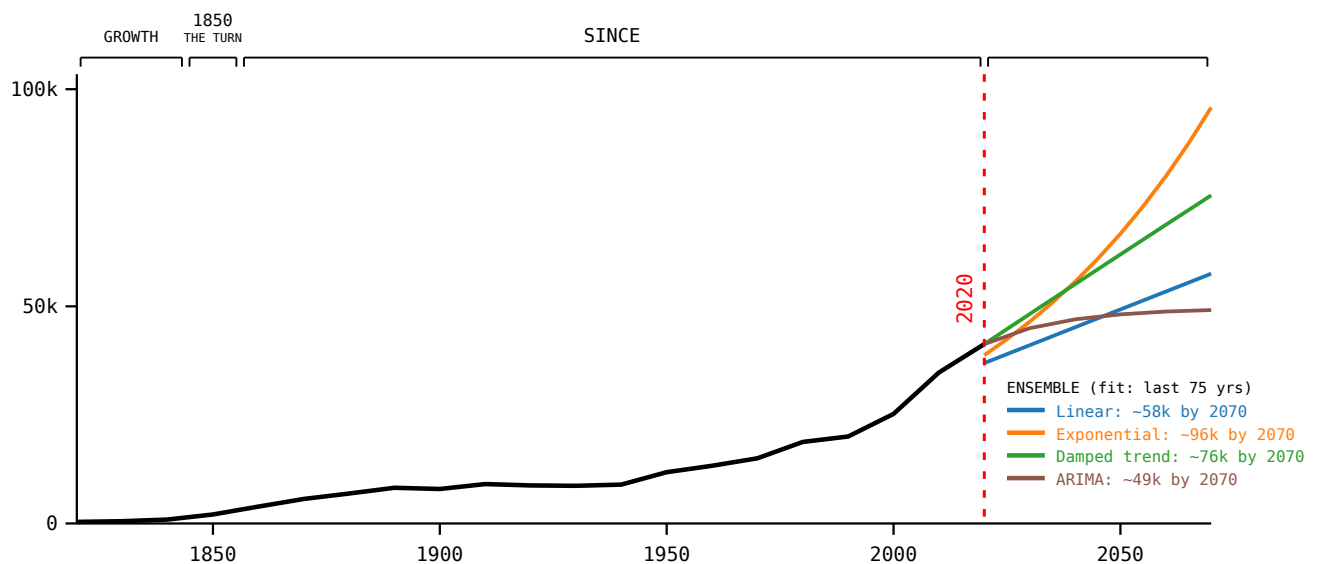


DELAWARE CO.

A quiet county seat and college town, birthplace of President Hayes, that after 2000 accelerates into a Columbus boomburb.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



project-artinscience • seed 1850 • edition _____ / _____ • signed _____

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Delaware

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VOLUME I — OHIO: TEN CITIES, 200 YEARS

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STD

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Delaware

Chapter 9 — Notes & Topics for Discussion (draft)

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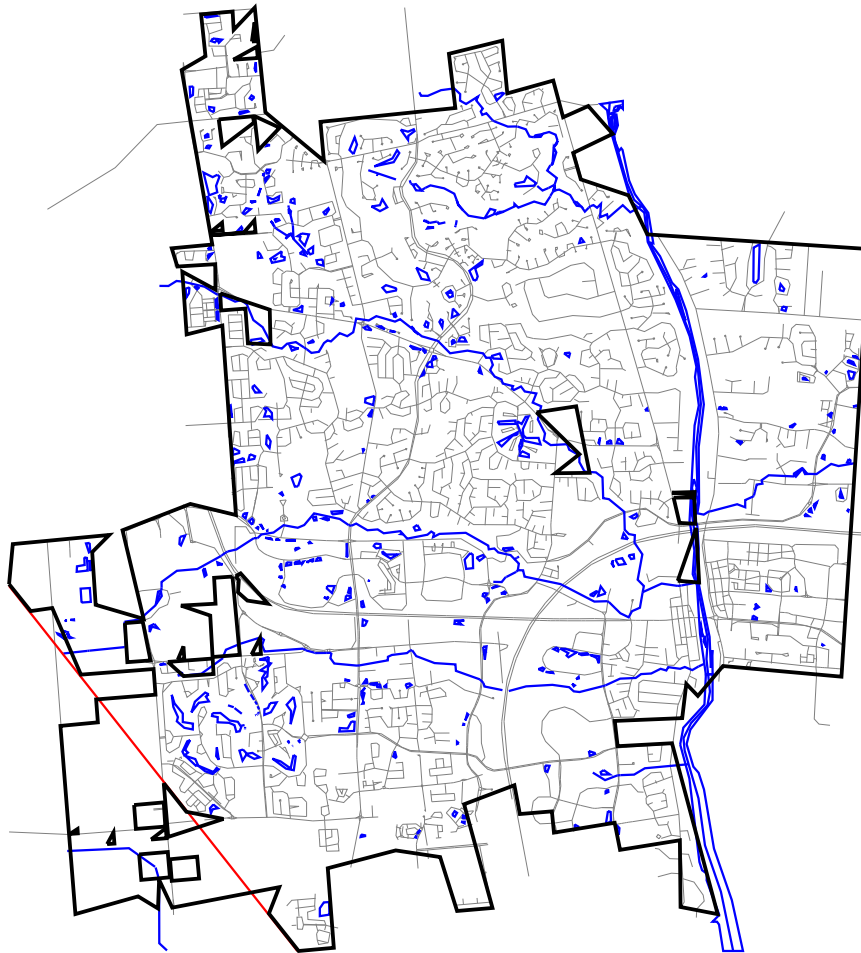
DUBLIN

Chapter 10 • 1820–2020, projected to 2120

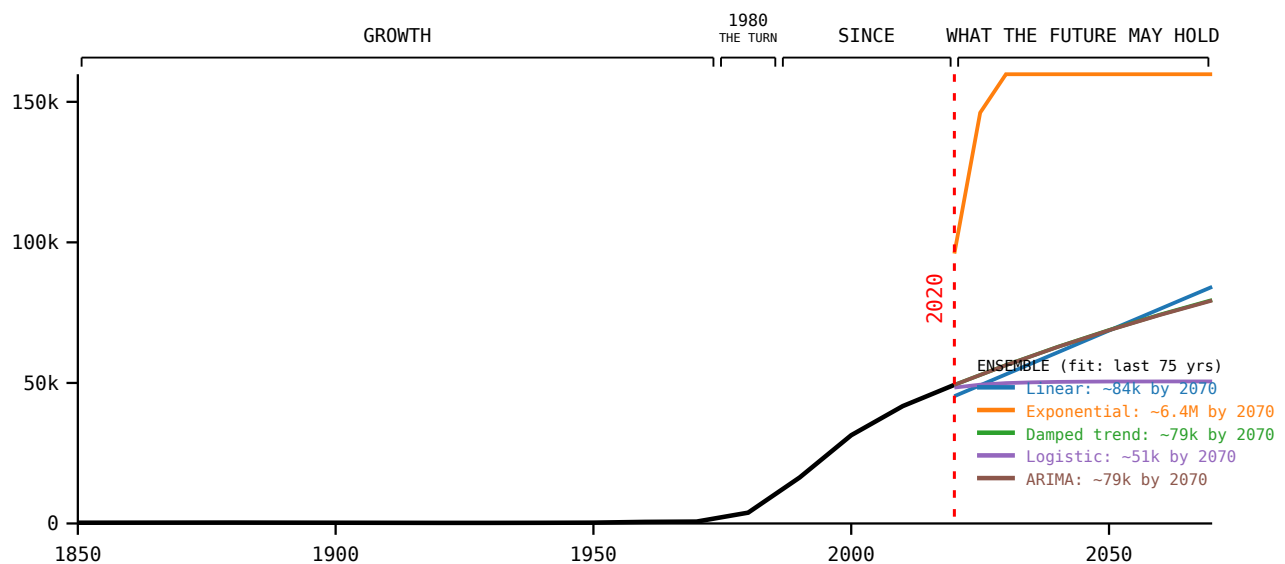


FRANKLIN CO.

Under 700 people in 1880, past 49,000 by 2020; no curve in the volume rises faster.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



project-artinscience • seed 1980 • edition _____ / _____ • signed _____

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Dublin

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VOLUME I — OHIO: TEN CITIES, 200 YEARS

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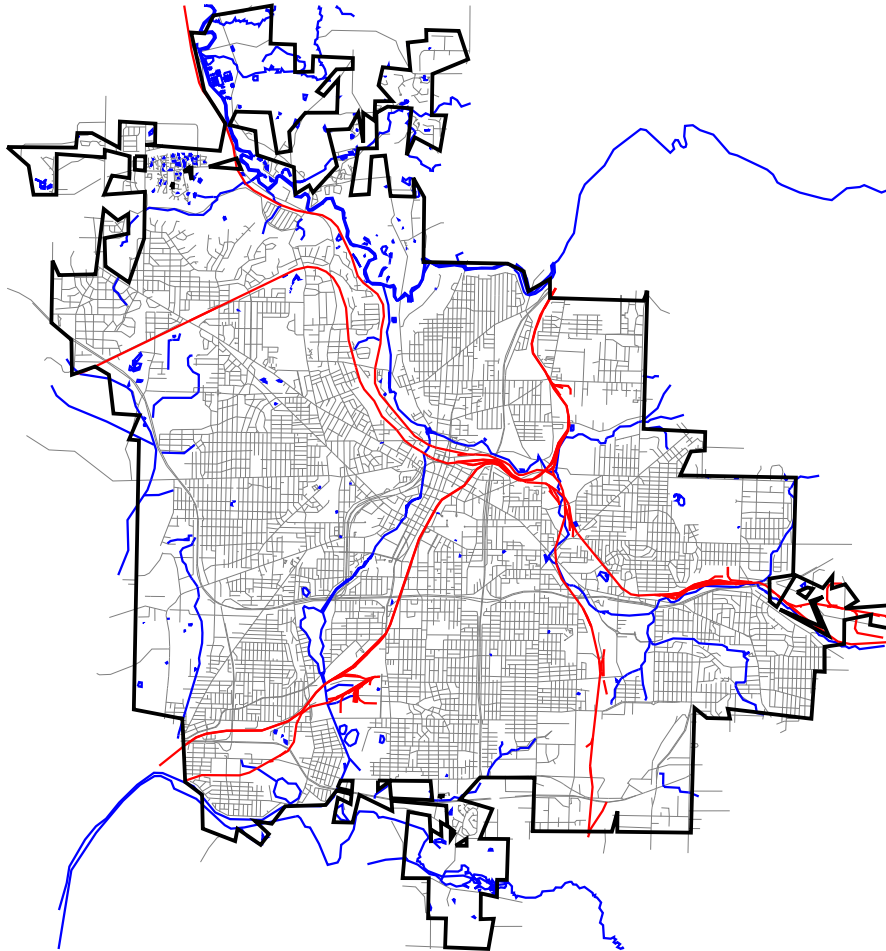
AKRON

Chapter 11 • 1820–2020, projected to 2120

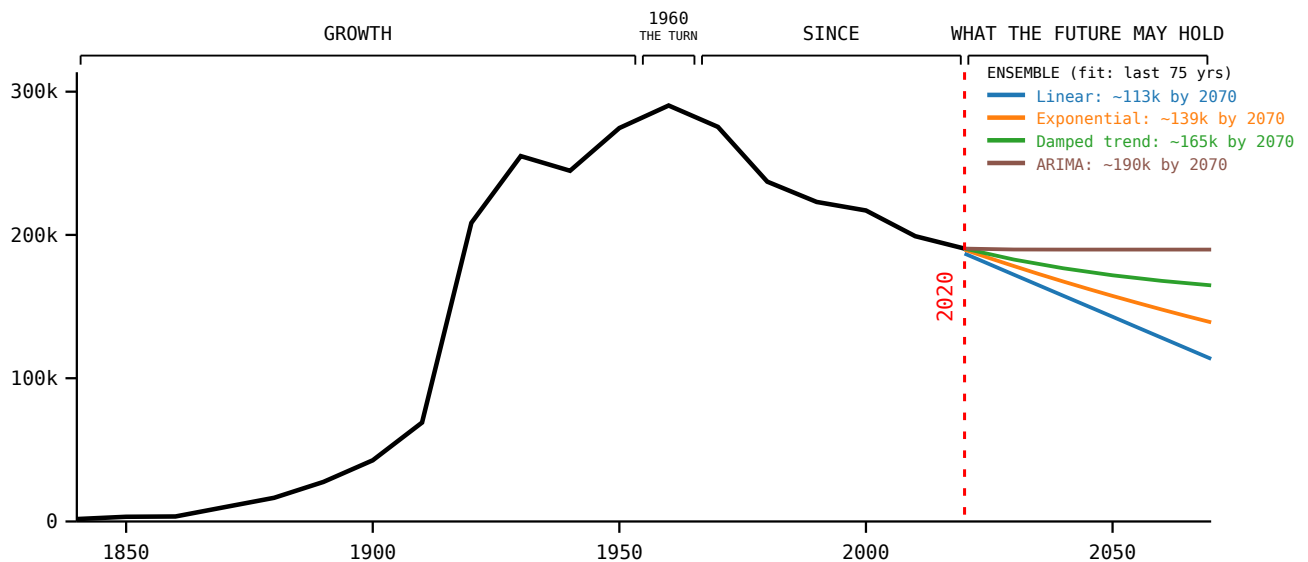


SUMMIT CO.

The “Rubber Capital of the World” peaked in 1960 at over 290,000; the tour closes where the industry — and the people — began their long slide.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



project-artinscience • seed 1960 • edition _____ / _____ • signed _____

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Akron

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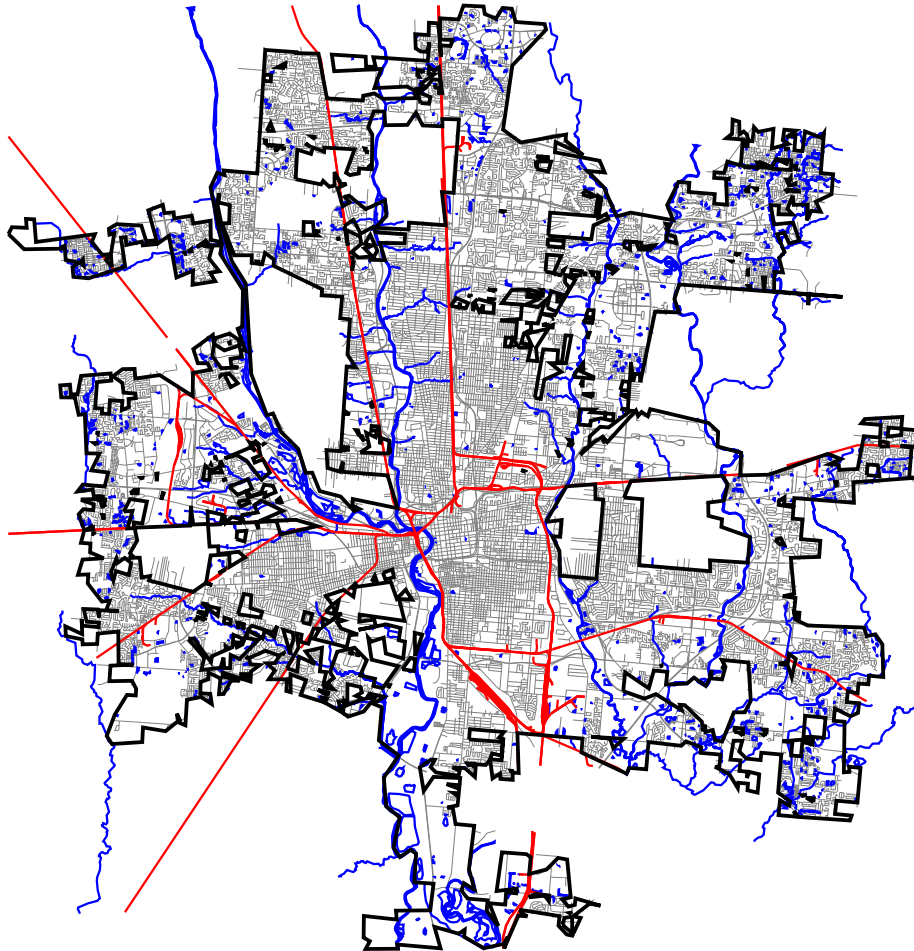
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Teaser • 1820–2020, projected to 2120

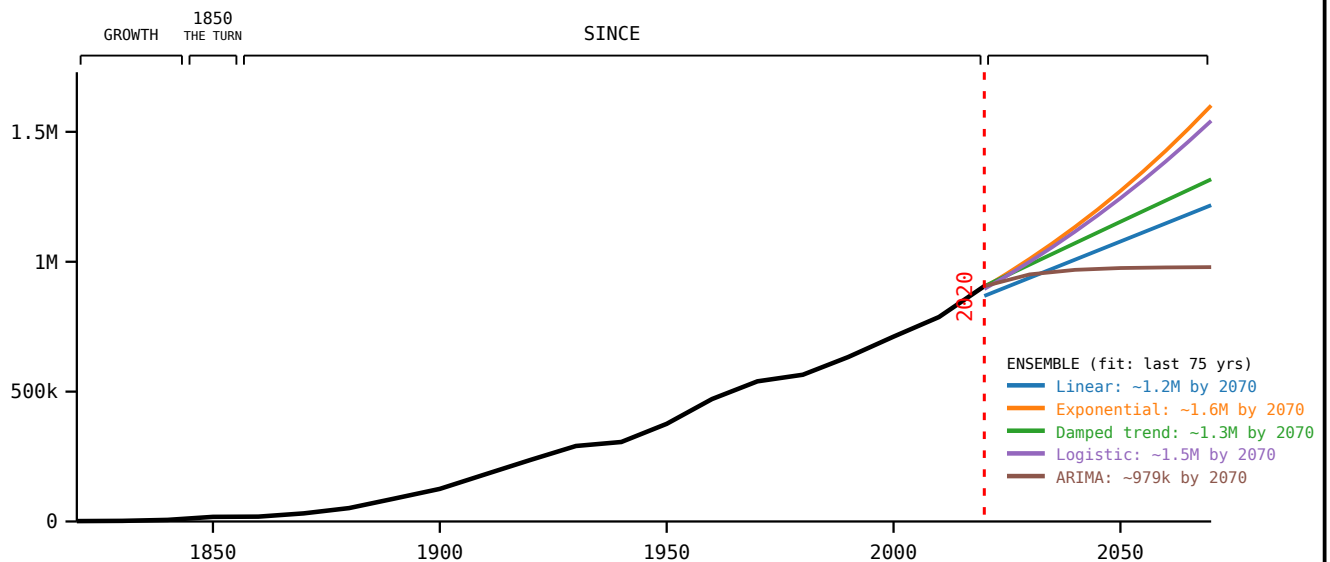


FRANKLIN CO.

The capital whose curve only ever climbs — a full-size teaser pointing past this volume.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



project-artinscience • seed 1850 • edition _____ / _____ • signed _____

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Columbus

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STC

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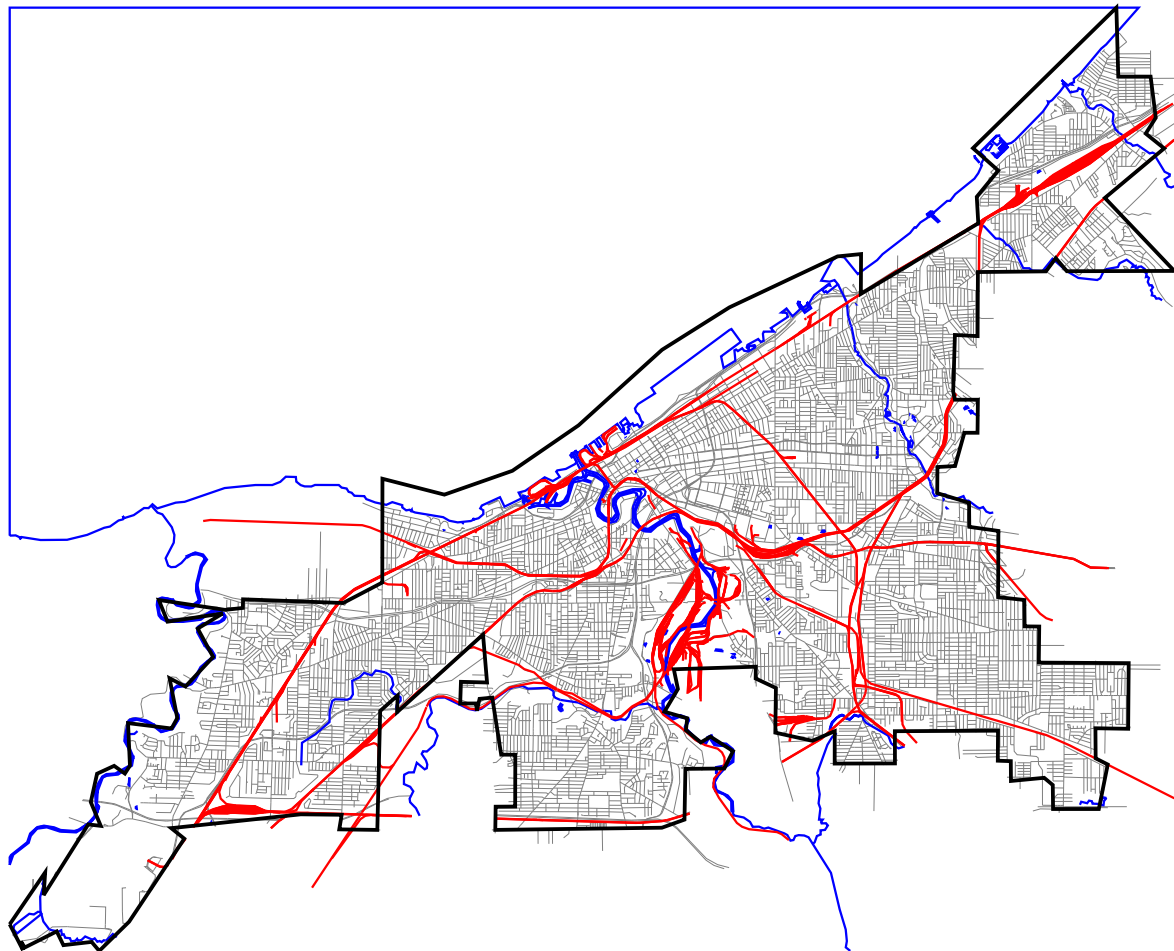
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Teaser • 1820–2020, projected to 2120

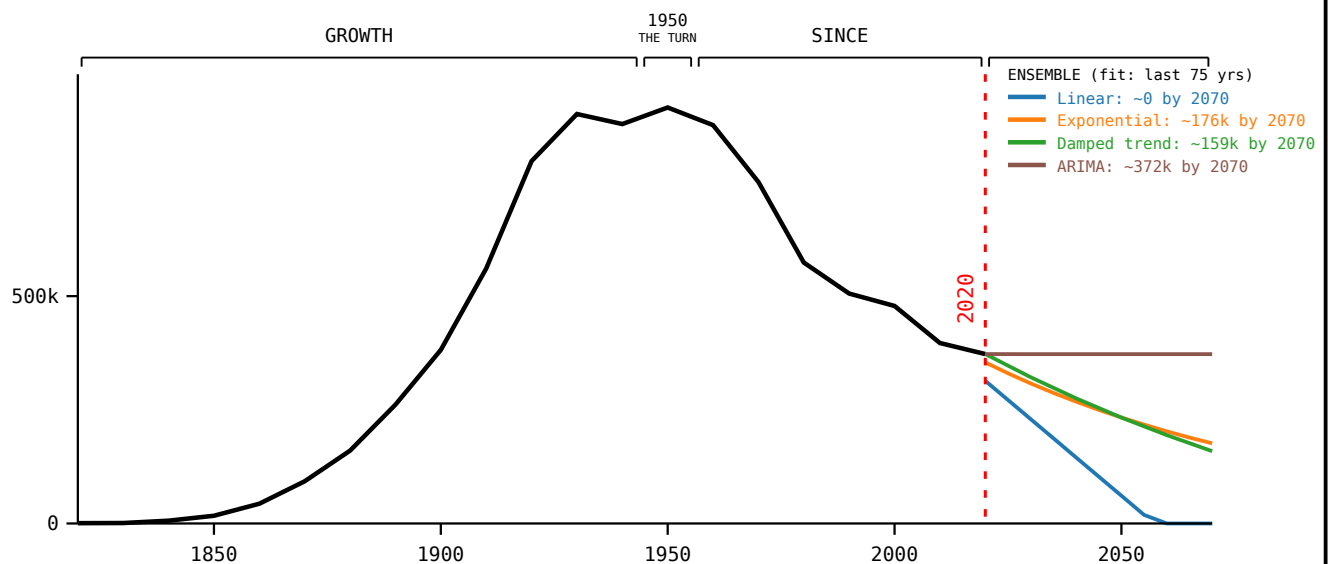


CUYAHOGA CO.

The big lakefront boom and bust at metropolitan scale — a teaser for another volume.



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project-artinscience • seed 1950 • edition _____ / _____ • signed _____

VOLUME I — OHIO: TEN CITIES, 200 YEARS

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$$P(t) = a t + b$$

2. Exponential — log-linear. Constant *percentage* growth: least squares on the log of population, giving a fixed annual rate r . Aggressive for fast-growing places.

$$\ln P(t) = a t + b \Rightarrow P(t) = P_0 e^{r(t-t_0)}$$

3. Damped trend — Holt exponential smoothing. A level ℓ_t and trend b_t re-estimated at each step, with the trend *damped* by $0 < \phi < 1$ so growth flattens over time.

$$\hat{P}_{t+h} = \ell_t + (\phi + \phi^2 + \dots + \phi^h) b_t$$

4. Logistic — saturating S-curve. Growth that slows toward a carrying capacity L ; contributes only where the data show an inflection.

$$P(t) = \frac{L}{1 + e^{-k(t-t_0)}}$$

5. ARIMA(1,1,0). A time-series model on the year-to-year *change* $\Delta P_t = P_t - P_{t-1}$, carrying momentum from the last differenced step.

$$\Delta P_t = \phi \Delta P_{t-1} + \varepsilon_t$$

A *cohort-component model* (census births, deaths and net migration) is the planned next ensemble member; capping the horizon at 50 years keeps the error bounded.

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VOLUME I — OHIO: TEN CITIES, 200 YEARS

Cleveland

Teaser — Data Sources & Software

Population — U.S. Decennial Census, 1820–2020.

What: the official decade-by-decade head count for the place. *Source:* U.S. Census Bureau, harmonised through IPUMS NHGIS (Minnesota Population Center) — which republishes consistent historical census tables and boundaries back to 1790. *How extracted:* the per-place decennial series was compiled into a plain `year, population` CSV and read directly by the engine. [Per-place figures to be spot-checked.]

Geography — OpenStreetMap.

What: the municipal boundary, roads, waterways and railways on the plate. *Source:* OpenStreetMap, a global, openly licensed, collaboratively edited geodatabase (© OpenStreetMap contributors, ODbL). *How extracted:* queried by place name with the `osmnx` library (Overpass API), cached locally as GeoJSON, then reprojected, simplified and clipped to the map's extent.

Projection / geodesy.

What: the coordinate systems the maps are drawn in. *Source:* the EPSG geodetic registry. *How applied:* Ohio maps use NAD83 / UTM zone 17N (EPSG:26917) so the state sits true north-up; the U.S. locator uses NAD83 / CONUS Albers (EPSG:5070). Reprojection via `geopandas / pyproj`.

Software & process.

artinscience Python engine (`shapely`, `geopandas`, `osmnx`, `statsmodels`, `scipy`); typeset in LaTeX (New Century Schoolbook); SVG rasterised with Inkscape; pen paths optimised with `vpype`; plotted on a UUNA TEK iDraw 2.0 (A4).

STD

Draft — full citations, retrieval dates and per-place verification to be finalised.

VOLUME I — OHIO: TEN CITIES, 200 YEARS

Cleveland

Teaser — Notes & Topics for Discussion (draft)

If you could time-travel, the year to visit

Also notable for

Population — detail, density & trend

Race & ethnicity ratios

Age & gender ratios

Tourism & attractions

Businesses & major employers

Crime & safety

Schools & education

Land use — residential / commercial / industrial zones

Nearby cities, towns & natural areas

Placeholders — to be researched and written in a later pass.

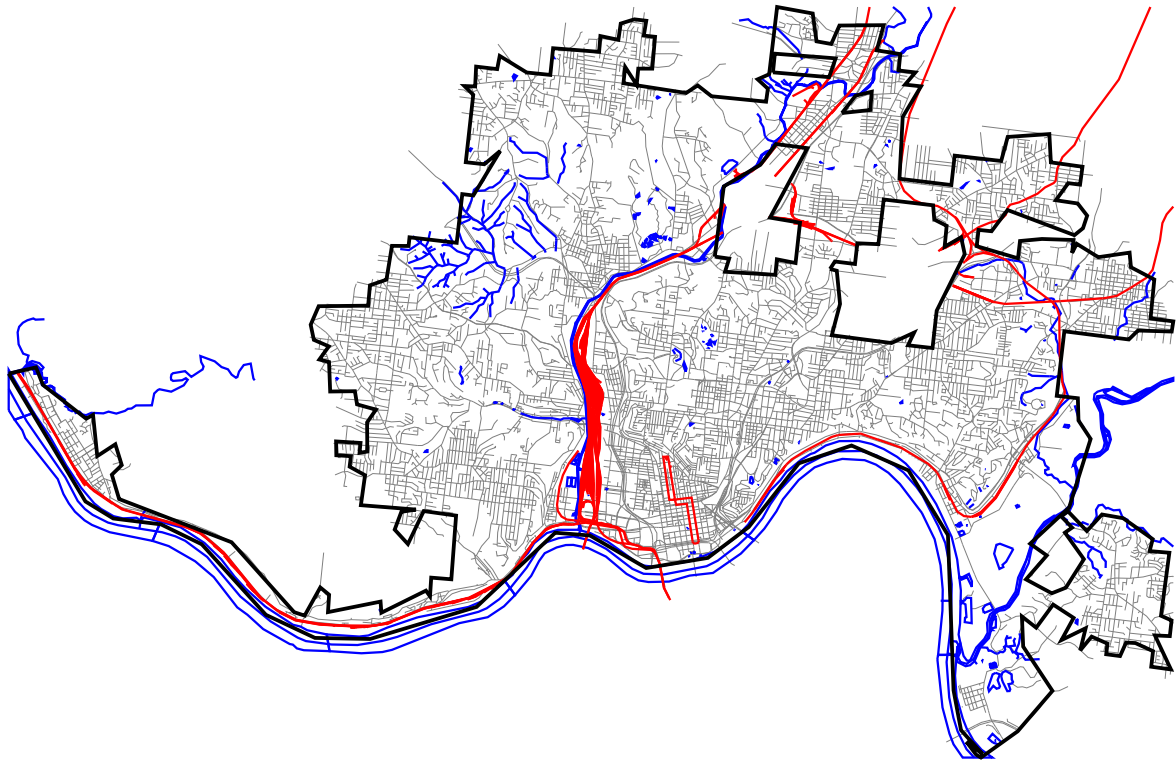
CINCINNATI

Teaser • 1820–2020, projected to 2120

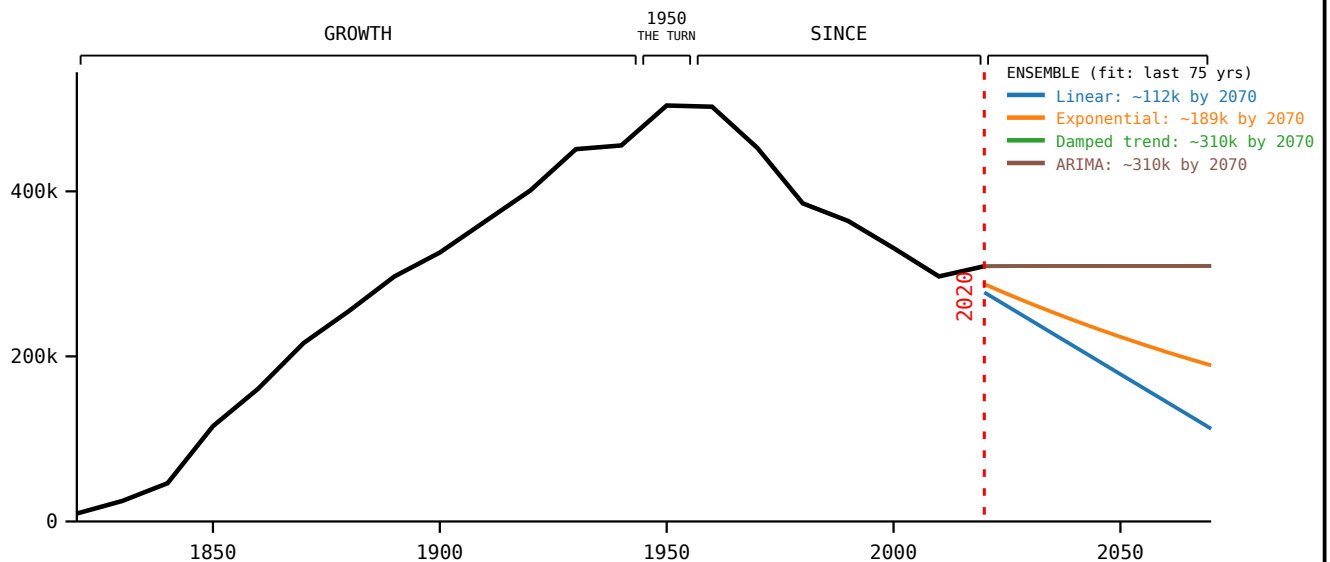


HAMILTON CO.

The nineteenth-century Queen City of the river, peaking early — the oldest arc of all.



LEGEND — City limits — Roads — Rivers / water — Railroads /// Highlighted county



project-artinscience • seed 1950 • edition _____ / _____ • signed _____

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Cincinnati

Teaser — Methodology: Forecast Algorithms

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